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Configuration Recommendations



For the full Pulse configuration refer to the official [Configuration Reference](#) and the recommended [Production Setup](#).

Setting up the JVM DNS cache TTL

A couple of security properties should be set up. This can be achieved by:

1. Making sure that the `security.overridePropertiesFile` property is set to true in the `$JAVA_HOME/jre/lib/security/java.security` file.
2. Creating a new file (e.g. `/opt/feedzai/pulse/conf/java.security.properties`) with the following content :

```
# conf/java.security.properties
networkaddress.cache.ttl=1
networkaddress.cache.negative.ttl=3
```

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1. Using the previously created file to set up the security properties on Pulse by adding `-Djava.security.properties=conf/java.security.properties` to the `JVM_OPTS` variable on `/opt/feedzai/pulse/bin/startup.sh` :

```
# /opt/feedzai/pulse/bin/startup.sh snippet
JVM_OPTS="(...) -Djava.security.properties=conf/java.security.properties"
```

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1. Using the previously created file to set up the security properties on App engines by adding `-Djava.security.properties=conf/java.security.properties` to the `pulse.global.appengine.jvmopts` property on Pulse properties.

```
# /opt/feedzai/pulse/conf/pulse.properties snippet
pulse.global.appengine.jvmopts=(...) -Djava.security.properties=conf/java.security.properties
```

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Setting up these properties affects the cache behavior of all network address name resolutions for both Pulse server and App engine processes.

From the `java.security` file, next to the `networkaddress.cache.ttl` property on, for example, Zulu JDK:

"NOTE: setting this to anything other than the default value can have serious security implications. Do not set it unless you are sure you are not exposed to DNS spoofing attack."

Ansible



Available ansible variables for running the ansible playbook.

Variable	Default Value	Description
ansible_user	feedzai	User used to execute ansible-playbook
ansible_local_user	{{ ansible_user }}	User name of local user who owns the unpatched assets in the system. The default value is the same as ansible_user.
pulse_install_path	/opt/feedzai	Pulse installation path
pulse_home_path	"{{ pulse_install_path }}/product-pulse-{{ pulse_version }}"	Pulse home directory
pulse_license_src_path	"{{ basesoftware_path }}/pulse-licenses/dev"	Source file location for Pulse licenses
pulse_spark_dist_archive_src_path	"{{ basesoftware_path }}/{{ spark_dist_archive }}"	Source file location for Spark distribution archive
pulse_sso_idp_cert		Pulse SSO IDP certificate
pulse_sso_private_key		Pulse SSO private key
pulse_sso_public_key		Pulse SSO public key
pulse_sso_force_https	false	Force HTTPS for SSO. If true, HTTP requests behind a proxy will require https:// and https:// is enforced.
pulse_api_url	"{{ jwk_service_host }}"	URL of the JWK service API, including protocol, host, port, and scheme. Example: http://localhost:8080 or https://example.com:443. <host> must be replaced by the actual host address. Note: Multiple endpoints are available for different purposes. See the JWK Service Balancing section for more details.
cm_sso_private_key		Case Manager SSO private key
cm_sso_public_key		Case Manager SSO public key
reference_data_servers	reference-data	Reference data servers
reference_data_port	"{{ reference_data_app_port }}"	Reference data server port. This port is used for connecting to the reference data server using the REST API.
zookeeper_nodes	zookeeper	ZooKeeper nodes
cassandra_nodes	"{{ groups['cassandra'] }}"	Cassandra nodes
messaging_brokers	"rabbitmq:5672"	Messaging brokers
keycloak_url	"http://{{ keycloak_hostname }}:{{ keycloak_port }}"	Keycloak URL
keycloak_hostname	"{{ groups['keycloak'][0] }}"	Keycloak hostname

Variable	Default Value	Description
keycloak_port	8080	Keycloak port
cassandra_pulse_keyspace_replication	"{ 'class': 'NetworkTopologyStrategy', 'dc1': 1 }"	Cassandra pulse keyspace replication strategy
cassandra_cluster_name	"{{ inventory }} cluster"	Cassandra cluster name
cassandra_pulse_keyspace	"profiles"	Cassandra pulse keyspace
cassandra_pulse_username	"pulse"	Cassandra pulse username
cassandra_pulse_password	"secret"	Cassandra pulse password
database_pulse_eventstorage_hostname	"postgres"	Default PostgreSQL event storage host
database_pulse_eventstorage_port	"5432"	Default PostgreSQL event storage port
database_pulse_eventstorage_database_name	"feedzai"	Default PostgreSQL event storage database name
database_pulse_metadata_hostname	"postgres"	Default PostgreSQL metadata host
database_pulse_metadata_port	"5432"	Default PostgreSQL metadata port
database_pulse_metadata_database_name	"feedzai"	Default PostgreSQL metadata database name
database_pulse_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_hostname }}"	Default PostgreSQL event storage read-only host
database_pulse_eventstorage_read_only_port	"{{ database_pulse_eventstorage_port }}"	Default PostgreSQL event storage read-only port
database_pulse_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_database_name }}"	Default PostgreSQL event storage read-only database name
database_pulse_metadata_jdbc	"jdbc:postgresql://{{ database_pulse_metadata_hostname }}:{{ database_pulse_metadata_port }}/{{ database_pulse_metadata_database_name }}?{{ database_pulse_metadata_jdbc_opts }}"	PostgreSQL metadata JDBC connection string
database_pulse_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_eventstorage_hostname }}:{{ database_pulse_eventstorage_port }}/{{ database_pulse_eventstorage_database_name }}?{{ database_pulse_eventstorage_jdbc_opts }}"	PostgreSQL event storage JDBC connection string
database_pulse_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_eventstorage_read_only_hostname }}:{{ database_pulse_eventstorage_read_only_port }}/{{ database_pulse_eventstorage_read_only_database_name }}?{{ database_pulse_eventstorage_jdbc_opts_read_only }}"	PostgreSQL event storage read-only JDBC connection string
database_pulse_eventstorage_jdbc_opts	[currentSchema={{ database_pulse_eventstorage_schema }}, cancelSignalTimeout=2, connectTimeout=2, loadBalanceHosts=true,	PostgreSQL event storage JDBC options

Variable	Default Value	Description
	loginTimeout=10, socketTimeout=60, tcpKeepAlive=true, targetServerType=master, reWriteBatchedInserts=true]	Database connection options
database_pulse_eventstorage_jdbc_opts_read_only	[currentSchema={{ database_pulse_eventstorage_schema }}, cancelSignalTimeout=2, connectTimeout=2, loadBalanceHosts=true, loginTimeout=10, socketTimeout=60, tcpKeepAlive=true, targetServerType={{ database_pulse_eventstorage_target_server_type_read_only }}]	PostgreSQL event storage database connection options
database_pulse_eventstorage_target_server_type_read_only	master	PostgreSQL event storage database target server type
jwk_service_db_jdbc	jdbc:postgresql://postgres:5432/feedzai?currentSchema=jwkservice	JDBC connection string for JWK service database
jwk_service_db_schema	jwkservice	Schema name for JWK service database
jwk_service_db_username	jwkservice	Username for JWK service database
jwk_service_db_password	secret	Password for JWK service database
jwk_service_host	"http://localhost:8080"	JWK service host address
rabbitmq_brokers	"{{ messaging_brokers }}"	RabbitMQ brokers list
rabbitmq_queue_type	"QUORUM"	RabbitMQ queue type
rabbitmq_pulse_username	"pulse"	RabbitMQ pulse queue username
rabbitmq_pulse_password	"secret"	RabbitMQ pulse queue password
rabbitmq_pulse_vhost	"pulse{{ deployment_color }}"	RabbitMQ pulse queue vhost
rabbitmq_pulse_queue_type	"QUORUM"	RabbitMQ pulse queue type
rabbitmq_feedback_storage_service_vhost	"feedback-storage"	RabbitMQ feedback storage service vhost
rabbitmq_casemanager_brokers	"{{ rabbitmq_brokers }}"	RabbitMQ casemanager brokers list
rabbitmq_casemanager_username	"cm"	RabbitMQ casemanager username

Variable	Default Value	Description
		Alert to send to Case Manager
rabbitmq_casemanager_password	"secret"	RabbitMQ password for Case Manager cluster
rabbitmq_casemanager_vhost	"cm"	RabbitMQ vhost used by Case Manager Storage and send to Case Manager
database_pulse_metadata_username	"pulse"	Pulse database username for metadata
database_pulse_metadata_password	"secret"	Pulse database password for metadata
database_pulse_metadata_schema	"pulse"	Pulse database schema for metadata
database_pulse_eventstorage_username	"eventstorage"	Pulse database username for eventstorage
database_pulse_eventstorage_password	"secret"	Pulse database password for eventstorage
database_pulse_eventstorage_schema	"eventstorage"	Pulse database schema for eventstorage
pulse_casemanager_username	casemanager	Case Manager username
pulse_casemanager_password	Secret1!	Case Manager password
pulse_batchprocessor_username	batchprocessor	Batch processor username
pulse_batchprocessor_password	Secret1!	Batch processor password
pull_hadoop_files_from_bucket	true	Pull data from Hadoop configuration bucket
hadoop_files_download_dir	"/tmp"	Temporary directory for downloading files
gcp_bucket_name	"pulse-dataproc-config"	The name of the GCP bucket
gcp_hadoop_conf_file	"hadoop-files.tar.gz"	Hadoop configuration file
model_training_directory	"hdfs:///user/pulse/models"	The directory where model training files should be stored. This is a local directory (HDFS).

Variable	Default Value	Description
model_storage_directory	"hdfs:///user/pulse/models"	The directory where models are stored. Can be local or HDFS.
model_hpt_directory	"hdfs:///user/pulse/hpt-models"	The directory where hyperparameter tuning models are stored. Can be local or HDFS.
number_threads_workflow_recovery	1	Number of threads used for workflow recovery.
pulse_spark_eventlog_enabled	false	Enable logging of Spark event logs.
pulse_spark_eventlog_dir	"hdfs:///var/log/spark/apps"	Directory for Spark event logs.
pulse_secret_value	6edf6aaf214a3b94b4536d1f5388d7ca	Secret value for Pulse.
pulse_properties_to_encrypt	[rabbitmq_pulse_password, rabbitmq_casemanager_password, database_pulse_metadata_password, database_pulse_eventstorage_password, jwk_service_db_password, cassandra_pulse_password, pulse_sso_private_key, cm_sso_private_key, pulse_user_password, dummy_token, jwk_service_client_password, reference_data_pulse_jwt_token, tokenizer_pulse_jwt_token]	List of properties to encrypt. Does not include sensitive information.
pulse_loggers	{}	Configuration for Pulse loggers.
spark_datascience_jdk_home		Path to the Java JDK used for Spark.
spark_distributed_jdk_home		Path to the Java JDK used for distributed Spark.

Variable	Default Value	Desc
		Sets ME f exec Data sche Jobs. defau will b

(*) The configuration files should be stored on a bucket in tar.gz format. The list of files inside the tar.gz should be:

- core-site.xml
- hdfs-site.xml
- mapred-site.xml
- yarn-site.xml

The service account used to fetch the configuration files from the bucket should have read permissions for that bucket (`storage.buckets.get` and `storage.buckets.list`).

Quartz Cron Scheduling🔗



For more details on Quartz Cron Scheduling expressions, refer to the [official Quartz CronTrigger Tutorial](#).

Field	Mandatory	Allowed Values	Description
Seconds	X	0-59	The cron expression for <code>second</code>
Minutes	X	0-59	The cron expression for <code>minute</code>
Hours	X	0-23	The cron expression for <code>hour</code>
Day of Month	X	1-31	The cron expression for <code>day of month</code> (**)
Month	X	1-12 or JAN-DEC	The cron expression for <code>month</code>
Day of Week	X	1-7 (1=SUN, 7=SAT) or SUN-SAT	The cron expression for <code>day of week</code> (**)
Year		blank, 1970-2099	The cron expression for <code>year</code>



(**) In Quartz, you must use the `?` character in either the Day of Month or Day of Week field to indicate "no specific value". You cannot specify both fields simultaneously, as they conflict!